Kernix 0.10.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

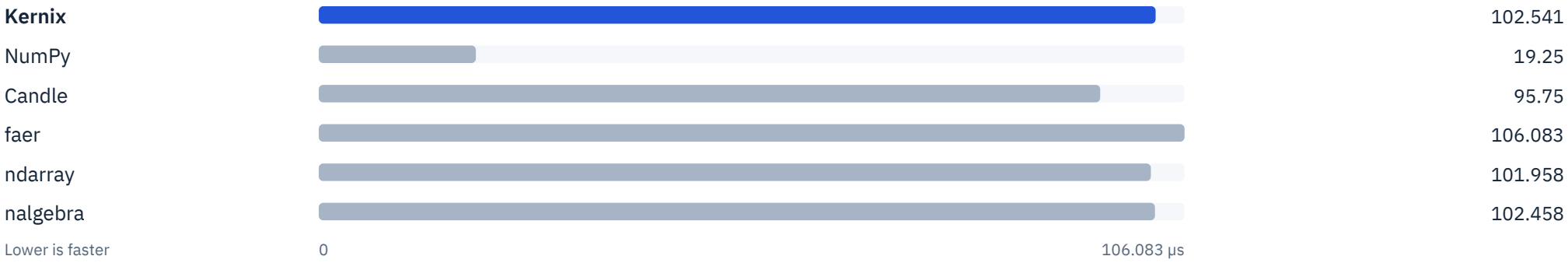
31 workloads

108 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

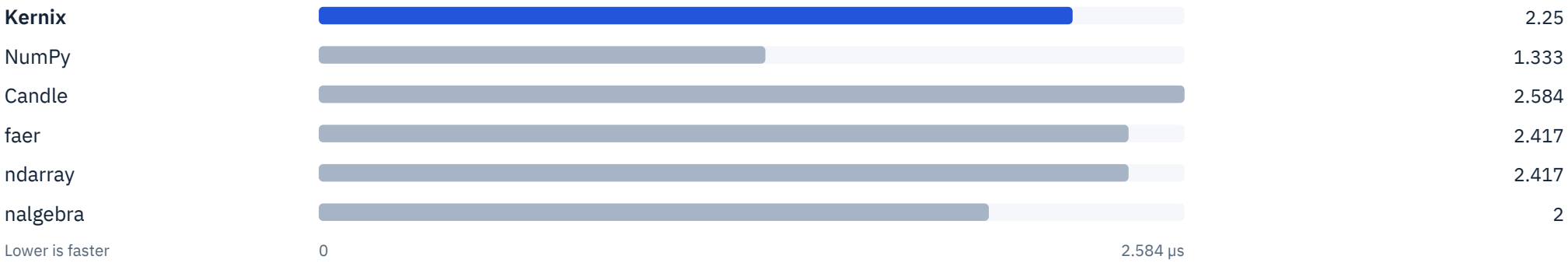
Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

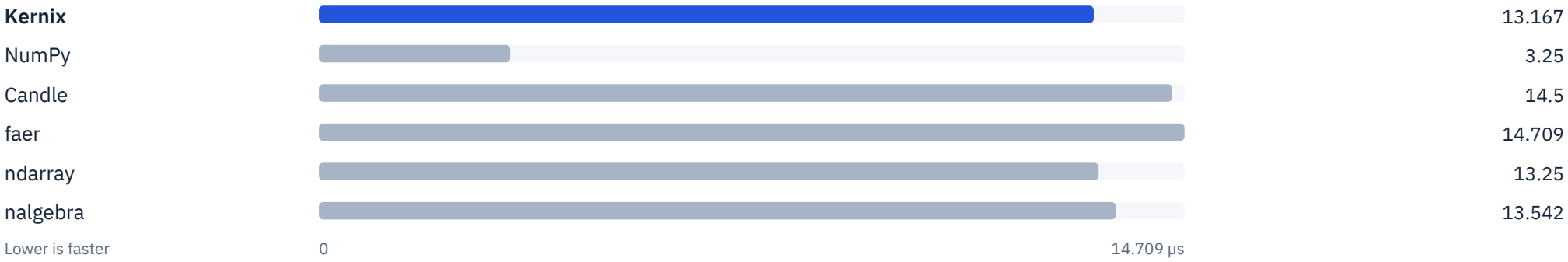
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



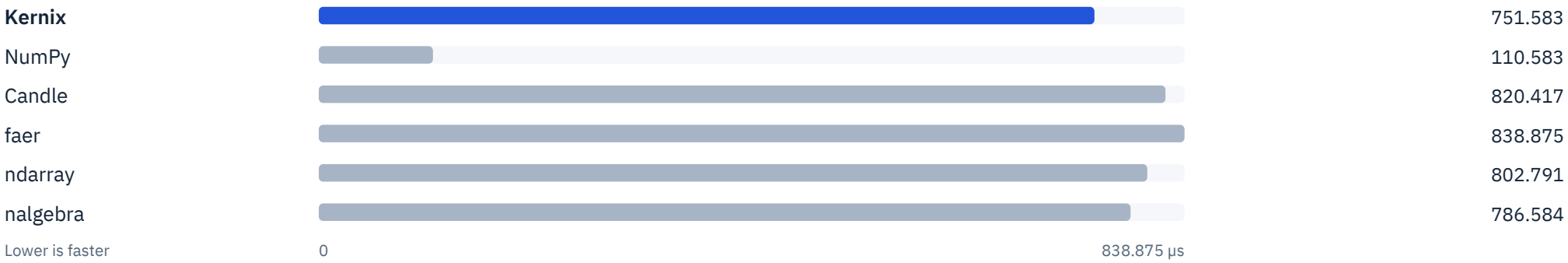
Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

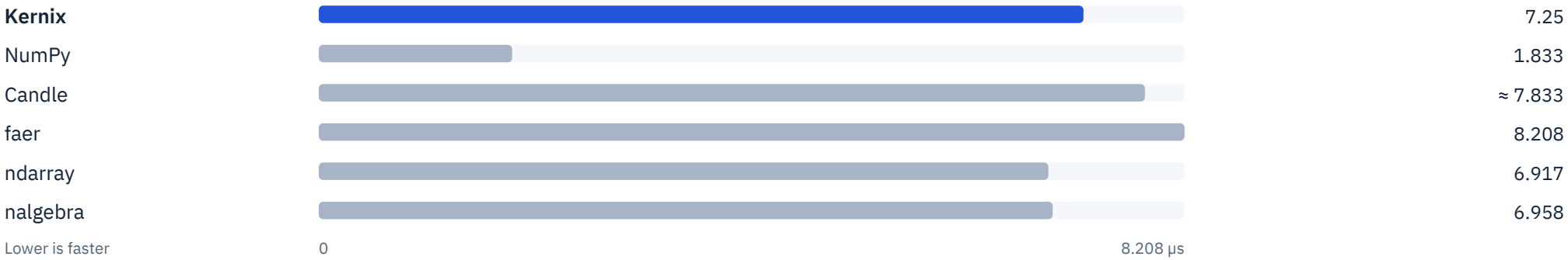
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



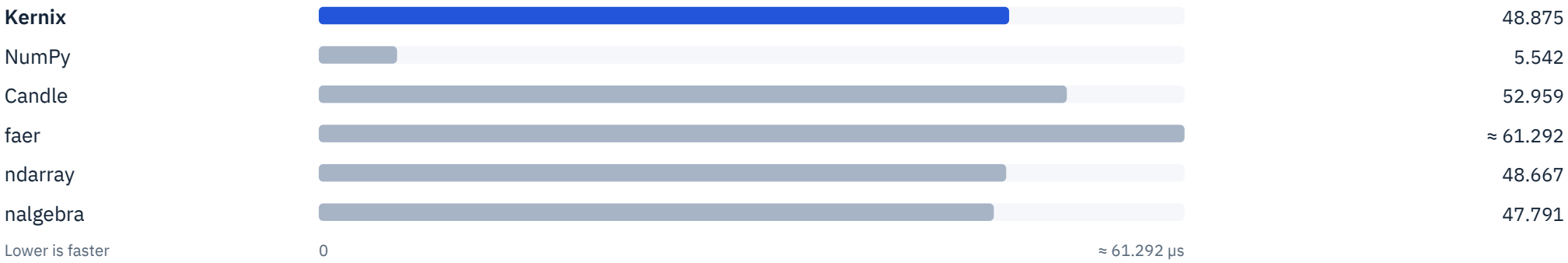
Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

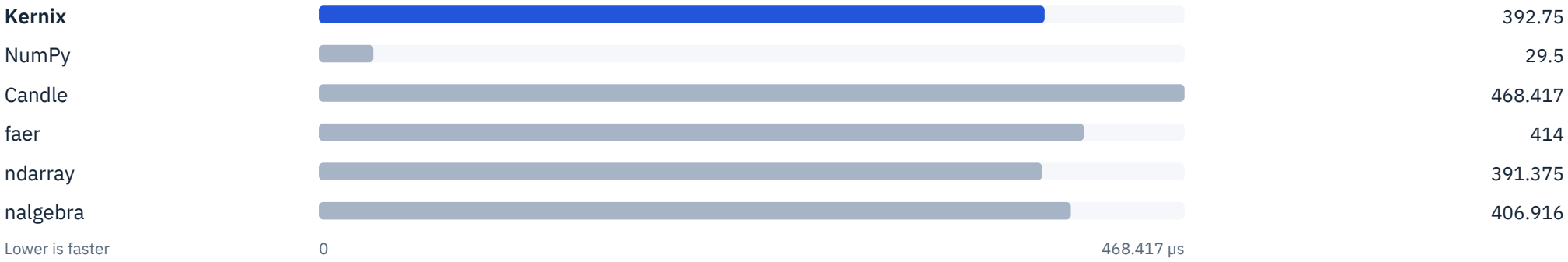
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

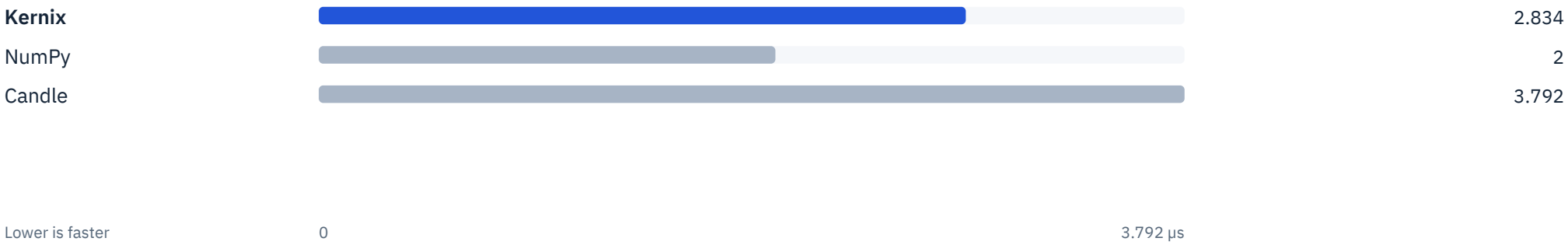


Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



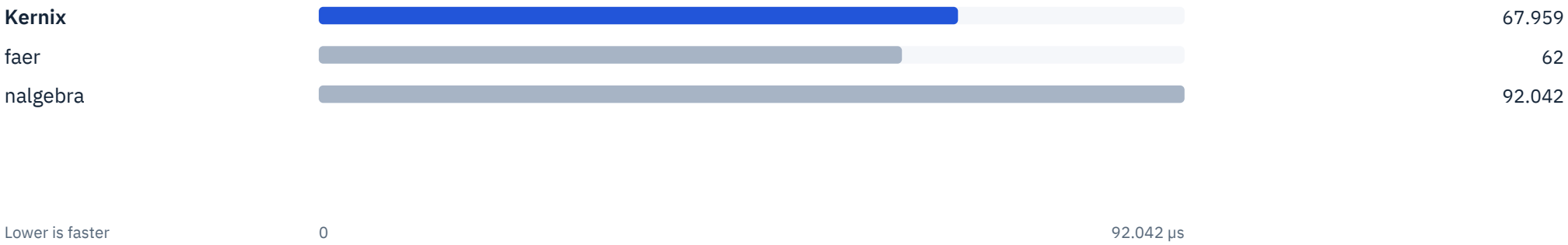
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

Per-call median

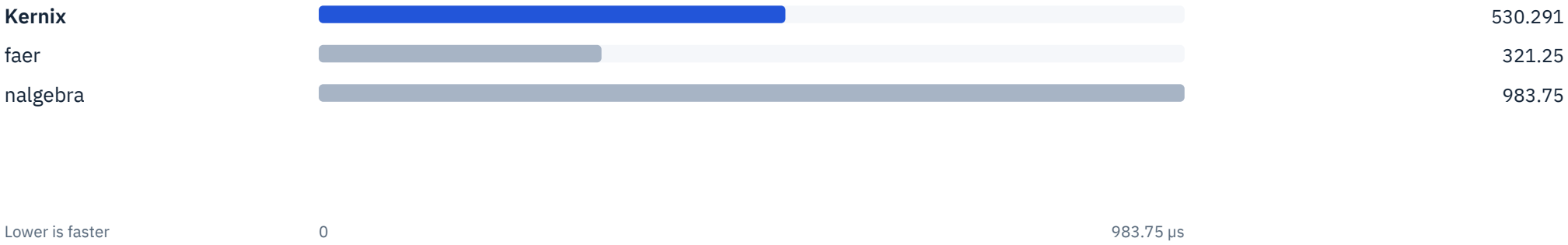
96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

256 × 64 · f64 · 1 worker · allocated output



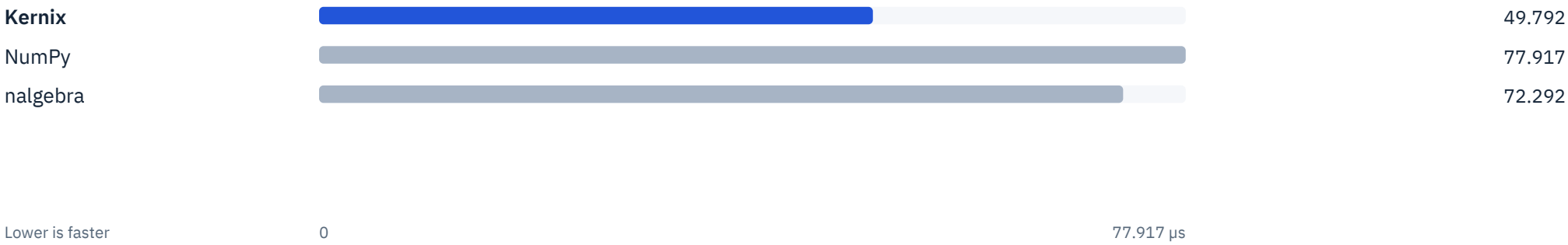
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

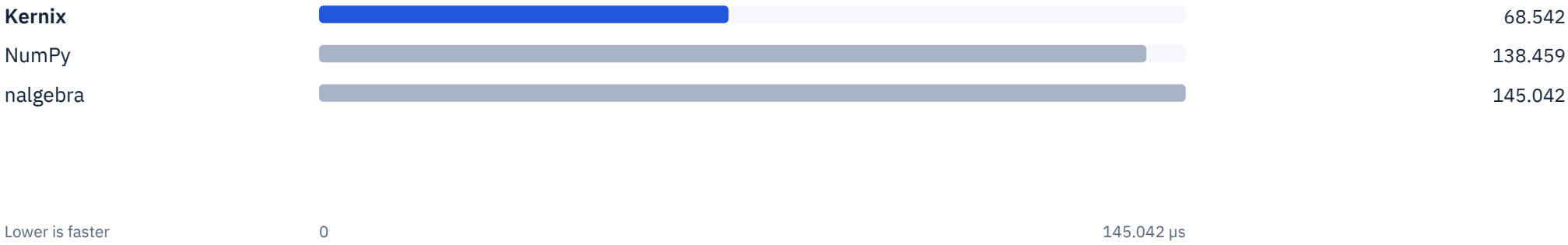
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



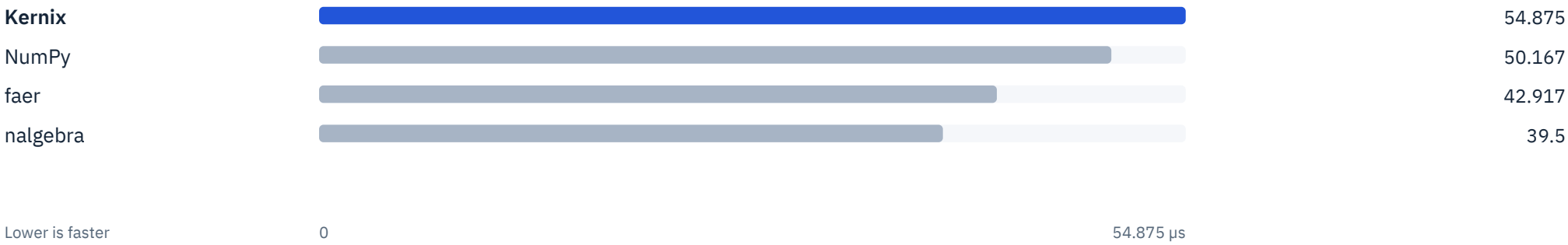
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

96 × 32 · f64 · 1 worker · allocated output

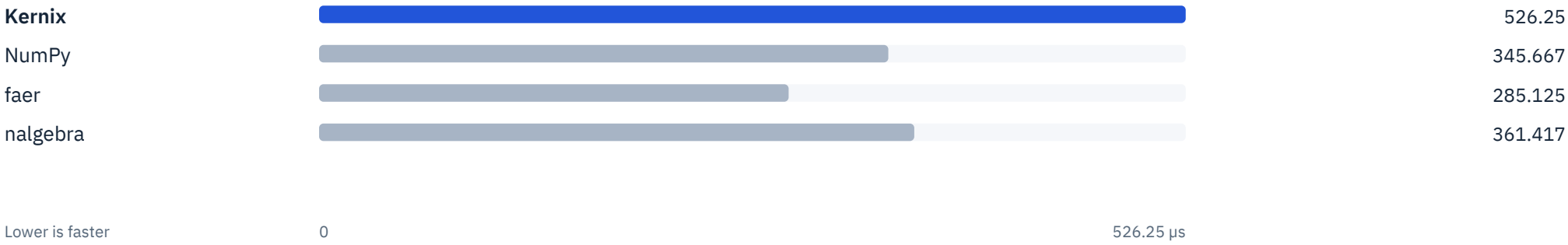
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



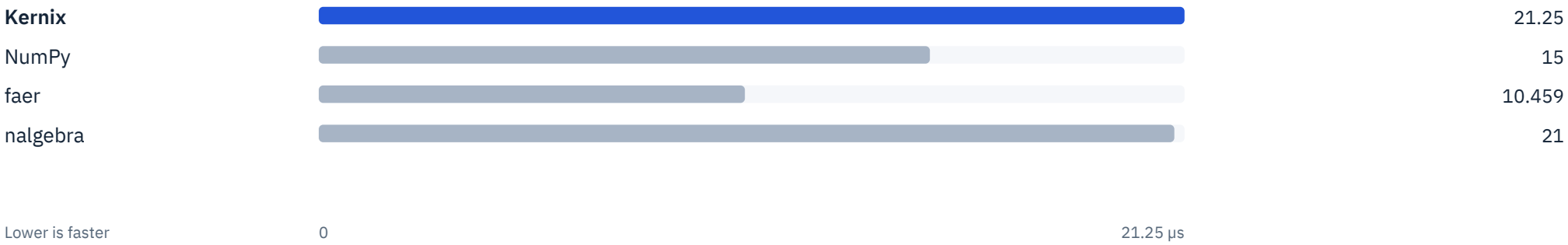
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

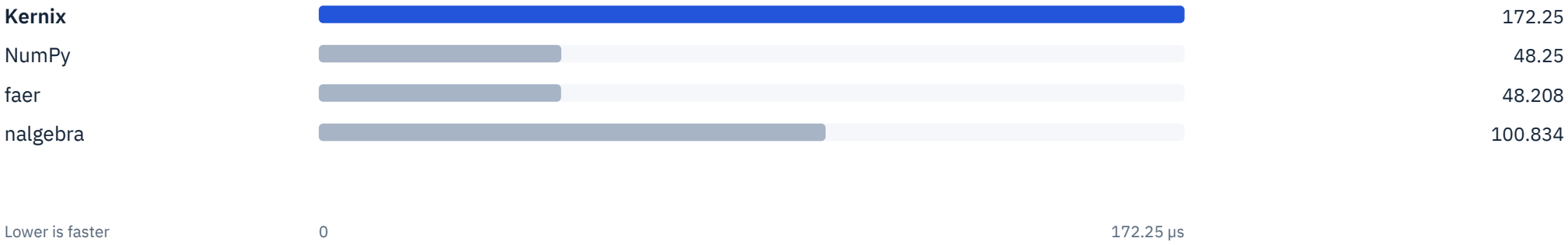
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



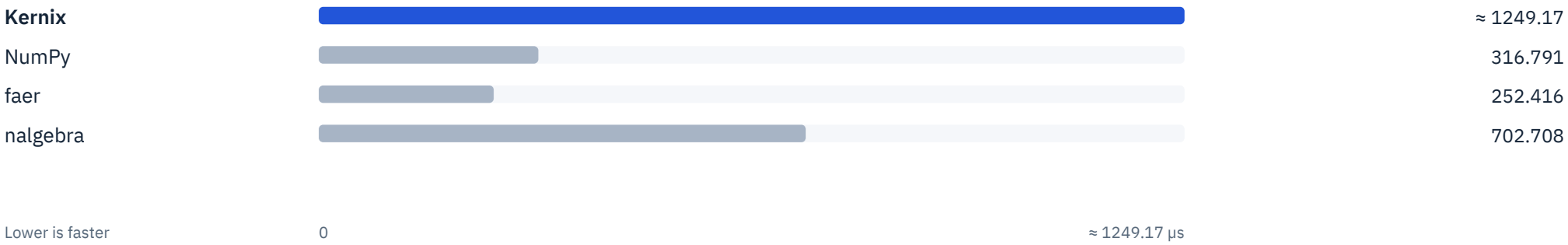
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



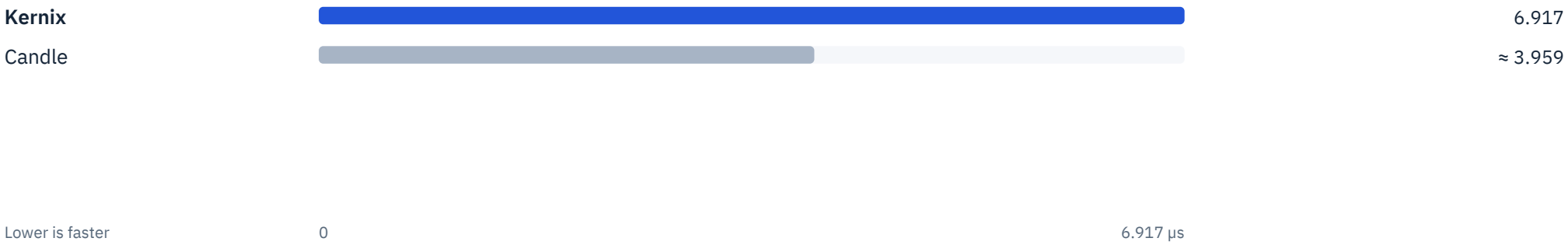
Kernix 0.10.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

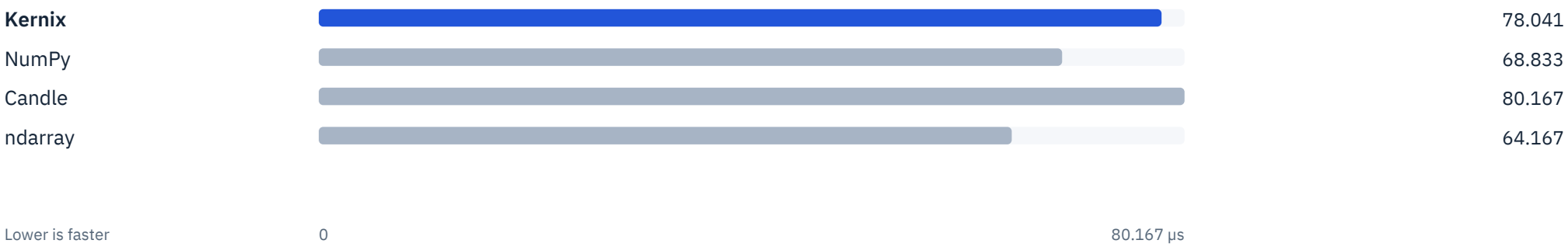


Kernix 0.10.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output

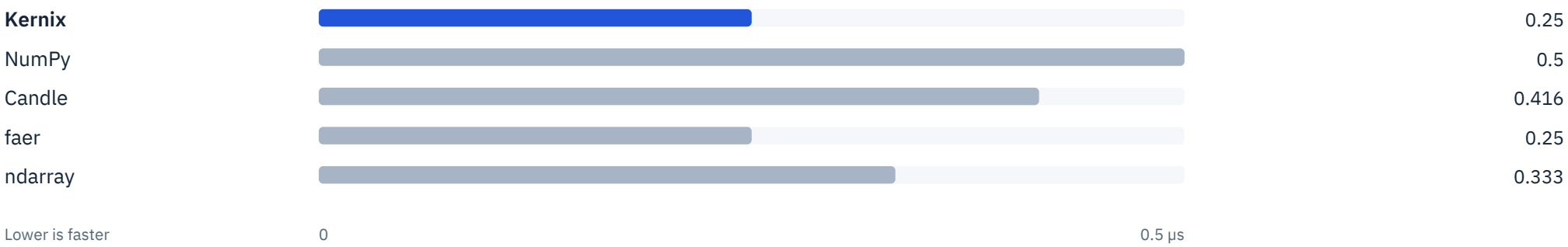


Kernix 0.10.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2584	2.584
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1333	1.333
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2000	2
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13250	13.25
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14709	14.709
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14500	14.5
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13167	13.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3250	3.25
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	13542	13.542
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	102541	102.541
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	106083	106.083
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	101958	101.958
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	102458	102.458
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	95750	95.75
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	19250	19.25
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	751583	751.583
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	110583	110.583

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	820417	820.417
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	838875	838.875
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	786584	786.584
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	802791	802.791
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6958	6.958
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7832.999999999999	7.832999999999999
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7250	7.25
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8208	8.208
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6917	6.917
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1833	1.833

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	61291.99999999999	61.29199999999999
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	5542	5.542
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	52959	52.959
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	48667	48.667
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	48875	48.875
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	47791	47.791
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	414000	414
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	392750	392.75
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	468417	468.417
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	406916	406.916

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	391375	391.375
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	29500	29.5
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3792	3.792
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2000	2
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2834	2.834
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	24250	24.25
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	89500	89.5
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	518917.00000000006	518.91700000000006
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	53500	53.5
Column-pivoted QR / Kernix 96 × 32 · f64	67959	67.959

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	62000	62
Column-pivoted QR / nalgebra 96 × 32 · f64	92042	92.042
Column-pivoted QR / Kernix 256 × 64 · f64	530291	530.291
Column-pivoted QR / faer 256 × 64 · f64	321250	321.25
Column-pivoted QR / nalgebra 256 × 64 · f64	983750	983.75
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	49792	49.792
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	72292	72.292
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	77917	77.917
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	145042	145.042
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	68542	68.542

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	138459	138.459
Thin QR / nalgebra 96 × 32 · f64	39500	39.5
Thin QR / NumPy 96 × 32 · f64	50167	50.167
Thin QR / Kernix 96 × 32 · f64	54875	54.875
Thin QR / faer 96 × 32 · f64	42917	42.917
Thin QR / faer 256 × 64 · f64	285125	285.125
Thin QR / Kernix 256 × 64 · f64	526250	526.25
Thin QR / NumPy 256 × 64 · f64	345667	345.667
Thin QR / nalgebra 256 × 64 · f64	361417	361.417
Cholesky factorization / NumPy 64 × 64 · f64	15000	15

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	21250	21.25
Cholesky factorization / faer 64 × 64 · f64	10459	10.459
Cholesky factorization / nalgebra 64 × 64 · f64	21000	21
Cholesky factorization / nalgebra 128 × 128 · f64	100834	100.834
Cholesky factorization / NumPy 128 × 128 · f64	48250	48.25
Cholesky factorization / faer 128 × 128 · f64	48208	48.208
Cholesky factorization / Kernix 128 × 128 · f64	172250	172.25
Cholesky factorization / nalgebra 256 × 256 · f64	702708	702.708
Cholesky factorization / faer 256 × 256 · f64	252416	252.416
Cholesky factorization / Kernix 256 × 256 · f64	1249167	1249.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	316791	316.791
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3958.9999999999995	3.9589999999999995
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6917	6.917
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6458	6.458
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2959	2.959
Model inference / Candle batched normalized/sliced MLP · f64	6208	6.208
Model inference / Kernix batched normalized/sliced MLP · f64	2333	2.333
Model recording + forward / Candle batched normalized/sliced MLP · f64	6125	6.125
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3000	3
Model backward / Kernix prepared graph → fresh gradients · f64	3542	3.542

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Candle prepared graph → fresh gradients · f64	20292	20.292
Parameter update / Kernix transactional SGD, fixed gradients · f64	708	0.708
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	61875	61.875
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5708	5.708
Row softmax / Candle 256 × 64 · f64	80167	80.167
Row softmax / ndarray 256 × 64 · f64	64167	64.167
Row softmax / Kernix 256 × 64 · f64	78041	78.041
Row softmax / NumPy 256 × 64 · f64	68833	68.833
Create zeros / Candle 64 × 64 · f32	416	0.416
Create zeros / faer 64 × 64 · f32	250	0.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	250	0.25
Create zeros / ndarray 64 × 64 · f32	333	0.333
Create zeros / NumPy 64 × 64 · f32	500	0.5
Create ones / Kernix 64 × 64 · f32	333	0.333
Create ones / Candle 64 × 64 · f32	375	0.375
Create ones / NumPy 64 × 64 · f32	1125	1.125
Create ones / ndarray 64 × 64 · f32	333	0.333
Create ones / faer 64 × 64 · f32	333	0.333